

SPARSEFIT: Few-shot Prompting with Sparse Fine-tuning for Jointly Generating Predictions and Natural Language Explanations

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Abstract

Models that generate natural language explanations (NLEs) for their predictions have recently gained increasing interest. However, this approach usually demands large datasets of human-written NLEs for the ground-truth answers at training time, which can be expensive and potentially infeasible for some applications. When only a few NLEs are available (a few-shot setup), fine-tuning pre-trained language models (PLMs) in conjunction with prompt-based learning has recently shown promising results. However, PLMs typically have billions of parameters, making full fine-tuning expensive. We propose SPARSEFIT, a sparse few-shot fine-tuning strategy that leverages discrete prompts to jointly generate predictions and NLEs. We experiment with SPARSEFIT on three sizes of the T5 language model and four datasets and compare it against existing state-of-the-art Parameter-Efficient Fine-Tuning (PEFT) techniques. We find that fine-tuning only 6.8% of the model parameters leads to competitive results for both the task performance and the quality of the generated NLEs compared to full fine-tuning of the model and produces better results on average than other PEFT methods in terms of predictive accuracy and NLE quality.

1 Introduction

Despite the tremendous success of neural networks (Chowdhery et al., 2022; Brown et al., 2020), these models usually lack human-intelligible explanations for their predictions, which are paramount for ensuring their trustworthiness. Building neural models that explain their predictions in natural language has seen increasing interest in recent years (Wiegrefe and Marasovic, 2021). Natural Language Explanations (NLEs) are generally easy to interpret by humans and more expressive than other types of explanations (Wallace et al., 2019; Wiegrefe and Marasovic, 2021). However, a significant downside of these models is that they require

large datasets of human-written NLEs at training time, which can be expensive and time-consuming to collect. To this end, few-shot learning of NLEs has recently emerged (Marasovic et al., 2022; Yordanov et al., 2022). However, current techniques involve fine-tuning the *entire* model with a few training NLE examples. This is computationally expensive since current NLE models can have billions of parameters (Schwartz et al., 2020).

In this paper, we investigate whether *sparse fine-tuning* (i.e. fine-tuning only a subset of parameters), in conjunction with prompt-based learning (i.e., textual instructions provided to a model (Liu et al., 2021)), can help in scenarios with limited availability of training instances with labels and NLEs. While sparse fine-tuning has been used in Natural Language Processing (NLP) (Houlsby et al., 2019; Logan et al., 2022; Zaken et al., 2022), to our knowledge, our work is the first to analyze sparse fine-tuning in the context of jointly generating predictions and NLEs. We extend the existing sparse fine-tuning strategy that looks only at bias terms (Zaken et al., 2022) to a comprehensive list of all layers and pairs of layers in a language model.

Thus, we propose SPARSEFIT, an efficient few-shot prompt-based training regime for models generating both predictions and NLEs for their predictions. We experiment with SPARSEFIT on two pre-trained language models (PLMs) that have previously shown high performance on task performance and NLE generation, namely T5 (Raffel et al., 2020) and UNIFIEDQA (Dong et al., 2019). We test our approach on four popular NLE datasets: e-SNLI (Camburu et al., 2018), ECQA (Aggarwal et al., 2021), SBIC (Sap et al., 2020), and ComVE (Wang et al., 2019), and evaluate both the task performance and the quality of the generated NLEs, the latter with both automatic metrics and human evaluation. Overall, SPARSEFIT shows competitive performance in few-shot learning settings with 48 training instances. For exam-

ple, fine-tuning only the Normalization Layer together with the Self-attention Query Layer, which amounts to 6.84% of the model’s parameters, consistently gave the best performance (penalized by the number of fine-tuned parameters) on both T5 and UNIFIEDQA over all four datasets. Remarkably, SPARSEFIT outperforms the current state-of-the-art parameter-efficient fine-tuning (PEFT) models in terms of both task performance and quality of generated NLEs in two of the four datasets. Furthermore, we find that fine-tuning other model components that comprise a small fraction of the parameters also consistently leads to competitive results; for instance, the *self-attention query* ($\sim 6.8\%$ of the parameters), the *self-attention query + LM head* ($\sim 11.3\%$), and the entire *self-attention layer* ($\sim 20\%$). Moreover, we also applied SPARSEFIT to larger language models (i.e. Llama 2–7B) and found that SPARSEFIT has competitive performance compared to the best PEFT strategy for all datasets. Therefore, we conclude that few-shot sparse fine-tuning of PLMs can achieve results competitive with fine-tuning the entire model.

2 Related Work

Few-shot learning refers to training models with limited labeled data for a given task (Finn et al., 2017; Vinyals et al., 2016). It has been successfully applied to several tasks such as image captioning (Dong et al., 2018), object classification (Ren et al., 2018), behavioral bio-metrics (Solano et al., 2020), graph node classification (Satorras and Estrach, 2018), and language modeling (Vinyals et al., 2016). Large Language Models (LLMs) have shown impressive skills to learn in few-shot scenarios (Brown et al., 2020; Chowdhery et al., 2022) thanks to the pre-training corpora size and the statistical capacity of the models (Izacard et al., 2022).

Parameter-Efficient Fine-Tuning Using fine-tuning, LLMs have shown breakthrough language understanding and generation capabilities in a wide range of domains (Raffel et al., 2020; Brown et al., 2020; Chowdhery et al., 2022). However, in NLP, the up-stream model (i.e., the model to be fine-tuned) is commonly a LLM with millions of parameters, such as T5 (Raffel et al., 2020), BERT (Devlin et al., 2019), or GPT-3 (Radford et al., 2018), which makes them computationally expensive to fine-tune. This has led to approaches known in the literature as Parameter-efficient Fine-tuning (PEFT) methods, which fine-tune only a small set

of the PLM’s parameters or an extra small set of parameters to keep competitive performance in the downstream task. In this regard, Li and Liang (2021) introduced *Prefix-Tuning*, a strategy that focuses on adding a small task-specific vector to the input so the frozen PLM can adapt its knowledge to further downstream tasks. Hu et al. (2022) developed *LoRA*, a technique that injects trainable low-rank matrices in the transformer architecture while freezing the pre-trained model weights. Zhang et al. (2023) extended this by proposing *AdaLoRA*, a method that adaptively allocates the rank budget among the low-rank matrices during training according to an importance score. Later, Zaken et al. (2022) presented BitFit, a novel approach aimed at only fine-tuning the bias terms in each layer of a transformer-based LM. We extend their work to fine-tuning some layers, or pairs of them, in the LM. More recently, Liu et al. (2022) introduced (*IA*)³, a fine-tuning method that scales the intermediate activations in a model with learned vectors.

Explainability of Neural Models Several approaches have been proposed in the literature to bring a degree of explainability to the predictions of neural models, using different forms of explanations, such as (1) Feature-based explanations (Ribeiro et al., 2016; Shrikumar et al., 2017; Yoon et al., 2019; Sha et al., 2021), (2) Natural Language Explanations (Camburu et al., 2018; Marasović et al., 2020; Kayser et al., 2022; Majumder et al., 2022), (3) Counterfactual explanations (Akula et al., 2020), and (4) Surrogate explanations (Alaa and van der Schaar, 2019). In this paper, we focus on models that provide NLEs, i.e., free-form text stating the reasons behind a prediction. Being in natural language, NLEs should be easy to interpret by humans and more expressive than other types of explanations, as they can present arguments that are not present in the input (Wiegraffe and Marasovic, 2021; Camburu et al., 2021; Kaur et al., 2020). NLEs have been applied to several domains such as question answering (Narang et al., 2020), natural language inference (Camburu et al., 2018), recommendation systems (Chen et al., 2021), reinforcement learning (Ehsan et al., 2018), medical imaging (Kayser et al., 2022), visual-textual reasoning (Hendricks et al., 2018; Kayser et al., 2021; Majumder et al., 2022), and solving mathematical problems (Ling et al., 2017).

To make neural models capable of generating accurate NLEs, the most common approach con-

sists of annotating predictions with human-written explanations and training models to generate the NLEs by casting them as a sequence generation task (Camburu et al., 2018). However, gathering large datasets with human-written NLEs is expensive and time-consuming. To address this, Yordanov et al. (2022) proposed a vanilla transfer learning strategy to learn from a few NLEs but abundant labels in a task by fine-tuning a PLM trained on a vast number of NLEs from other domains. More recently, Marasovic et al. (2022) introduced the FEB benchmark for few-shot learning of NLEs and a prompt-based fine-tuning strategy, which we use as a baseline in our work.

3 SPARSEFIT

We propose SPARSEFIT, an efficient few-shot NLE training strategy that focuses on fine-tuning only a subset of parameters in a large LM. SPARSEFIT is inspired by (1) Marasovic et al. (2022), who used fine-tuning and prompts to do few-shot learning of labels and NLEs; and (2) BitFit Zaken et al. (2022), who showed that fine-tuning only the bias terms in a PLM leads to competitive (and sometimes better) performance than fine-tuning the entire model. We extend BitFit by exploring the fine-tuning of different components (i.e., layers or blocks) in the PLM’s architecture. In particular, we study the self-rationalization performance after fine-tuning the following components in the T5 model: (1) the encoder blocks, (2) the decoder blocks, (3) the language model head, (4) the self-attention layers, (5) the feed-forward networks, (6) the normalization layer, and (7) all pairs of the above components that do not contain the encoder and decoder (see Appendix C). Given that UNIFIEDQA model’s architecture is the same as the one in T5, the interpretation of active parameters holds for UNIFIEDQA.

We aim to identify a set of guidelines for identifying which components should be fine-tuned to achieve competitive performance while updating a minimum number of parameters. Notice that when fine-tuning any component, or pair of components, we freeze all other PLM’s parameters and train the LM to conditionally generate a text in the form of “[label] because [explanation]”.

Encoder The T5 encoder comprises N transformer blocks, each composed of three layers: self-attention, position-wise fully connected layer, and layer normalization. The number of blocks depends

on the T5 variant (12 blocks for T5-base, 24 for T5-large, and 36 for T5-3B). The encoder accounts for roughly 41% of the model parameters.

Decoder The decoder accounts for roughly 54% of T5 model parameters. In addition to the self-attention, fully connected layer, and layer normalization, it also includes an encoder-decoder attention layer in its blocks, which we fine-tune as part of fine-tuning the decoder.

LM Head On top of the decoder, T5 has a language modeling head for generating text based on the corpus. The LM head accounts for roughly 5% of total model parameters.

Attention Layer Each of the transformer blocks starts with a self-attention layer. There are three types of parameters in the self-attention layer, namely, for computing the *query matrix* Q , the *key matrix* K , and the *value matrix* V . We propose to explore the fine-tuning of each self-attention matrix as a possible SPARSEFIT configuration. We also explore fine-tuning the *entire Self-attention Layer* (Q , K , V). On average, the percentage of trainable parameters associated with each matrix accounts for roughly 6% of model parameters. Note that the attention parameters in the encoder-decoder attention are not updated in this setting (they are only updated together with the decoder).

Layer Normalization The normalization layers are intended to improve the training speed of the models (Ba et al., 2016). The T5 model includes two *Layer Normalization* components per block, one after the self-attention layer and one after the feed-forwards networks. Unlike the original paper for layer normalization (Ba et al., 2016), the T5 model uses a simplified version of the layer normalization that only re-scales the activations. The percentage of learnable weights in the layer normalization is roughly 0.2% of the parameters.

4 Experiments

Datasets We follow the FEB benchmark for few-shot learning of NLEs (Marasovic et al., 2022) and consider four NLE datasets: e-SNLI for natural language inference (Camburu et al., 2018), ECQA for multiple-choice question answering (Aggarwal et al., 2021), ComVE for commonsense classification (Wang et al., 2019), and SBIC for offensiveness classification (Sap et al., 2020).

SPARSEFIT		ComVE	ECQA	SBIC	e-SNLI	Avg
Baseline (100.00%)	Acc.	80.5 ± 4.5	57.6 ± 2.6	70.1 ± 3.4	84.8 ± 2.6	73.3 ± 3.3
	nBERTs	74.2 ± 4.2	51.7 ± 2.4	67.8 ± 3.3	76.9 ± 2.5	67.7 ± 3.1
Decoder (54.60%)	Acc.	67.3 ± 6.0 ∇	58.5 ± 2.6	66.8 ± 3.1 ∇	86.6 ± 1.7 ∇	69.8 ± 3.4
	nBERTs	61.7 ± 5.5 ∇	52.3 ± 2.4 ∇	64.7 ± 2.7	78.3 ± 1.6 ∇	64.3 ± 3.0
Encoder (40.95%)	Acc.	72.6 ± 6.1 ∇	53.2 ± 3.6 ∇	62.4 ± 6.5 ∇	79.0 ± 3.4 ∇	66.8 ± 4.9
	nBERTs	67.1 ± 5.7	47.2 ± 3.2 ∇	58.7 ± 6.5 ∇	72.4 ± 3.2 ∇	61.3 ± 4.6
Dense.wo (27.29%)	Acc.	61.3 ± 4.4 ∇	56.1 ± 2.1 ∇	62.4 ± 2.6 ∇	84.0 ± 1.9	65.9 ± 2.8
	nBERTs	56.4 ± 4.1 ∇	0.0 ± 0.0 ∇	59.8 ± 2.6 ∇	74.7 ± 2.6 ∇	47.7 ± 2.3
Self-attention (KQV) (20.47%)	Acc.	76.2 ± 4.4 ∇	56.9 ± 3.0	69.9 ± 3.8	83.3 ± 2.4 ∇	71.6 ± 3.4
	nBERTs	70.3 ± 4.0 ∇	50.2 ± 2.7 ∇	67.4 ± 3.9 ∇	76.1 ± 2.2 ∇	66.0 ± 3.2
LM head + Attention.Q (11.28%)	Acc.	74.8 ± 5.0 ∇	55.4 ± 2.7 ∇	67.1 ± 5.2 ∇	82.8 ± 3.0 ∇	70.0 ± 4.0
	nBERTs	69.0 ± 4.6	43.7 ± 4.3 ∇	64.5 ± 5.5	75.8 ± 2.8 ∇	63.2 ± 4.3
LM head (4.46%)	Acc.	15.6 ± 1.3 ∇	58.9 ± 2.3 ∇	0.2 ± 0.2 ∇	86.7 ± 1.8 ∇	40.3 ± 1.4
	nBERTs	0.0 ± 0.0 ∇	0.0 ± 0.0 ∇	0.2 ± 0.2 ∇	0.0 ± 0.0 ∇	0.0 ± 0.0
LayerNorm + Attention.Q (6.84%)	Acc.	74.9 ± 5.3 ∇	55.8 ± 3.1 ∇	67.0 ± 4.4 ∇	82.6 ± 2.7 ∇	70.1 ± 3.9
	nBERTs	69.0 ± 4.8	45.9 ± 3.7 ∇	64.3 ± 4.7	75.6 ± 2.5 ∇	63.7 ± 3.9
Attention.K (6.82%)	Acc.	48.8 ± 2.8 ∇	56.7 ± 2.5 ∇	0.2 ± 0.2 ∇	19.6 ± 11.5 ∇	31.3 ± 4.3
	nBERTs	19.4 ± 10.0 ∇	0.0 ± 0.0 ∇	0.1 ± 0.2 ∇	0.2 ± 0.3 ∇	4.9 ± 2.6
Attention.Q (6.82%)	Acc.	74.8 ± 5.1 ∇	55.5 ± 3.2 ∇	66.9 ± 4.6 ∇	82.8 ± 2.6 ∇	70.0 ± 3.8
	nBERTs	68.9 ± 4.7	43.4 ± 4.8 ∇	64.2 ± 4.8	75.8 ± 2.3 ∇	63.1 ± 4.2
Attention.V (6.82%)	Acc.	55.5 ± 3.0 ∇	53.1 ± 2.8 ∇	30.1 ± 10.2 ∇	84.2 ± 2.0	55.7 ± 4.5
	nBERTs	51.0 ± 2.8 ∇	0.0 ± 0.0 ∇	30.1 ± 10.2 ∇	71.7 ± 3.4 ∇	38.2 ± 4.1
LayerNorm (0.02%)	Acc.	34.3 ± 2.4 ∇	59.0 ± 2.4 ∇	0.3 ± 0.3 ∇	86.6 ± 1.8 ∇	45.0 ± 1.7
	nBERTs	0.0 ± 0.0 ∇	0.0 ± 0.0 ∇	0.2 ± 0.2 ∇	0.0 ± 0.0 ∇	0.1 ± 0.1

Table 1: Summary of best performing SPARSEFIT configurations for T5-large. We report the average and the standard deviation over the 60 few-shot train-validation splits for the **accuracy** metric and the normalized BERTScore (**nBERTs**). In brackets are the percentages of fine-tuned weights for each SPARSEFIT configuration. We show in **bold** the setting with the highest metric for each dataset, in **blue** the highest performance among SPARSEFIT without considering the number of parameters, and in **green** the best-performing setting after considering the percentage of fine-tuned parameters. The trade-off between parameters and performances was computed using $(1 - \text{\%params}) \times \text{nBERTs}$. Significance testing was assessed via mean t-test compared with the baseline: ∇ represents a p-value lower than 10^{-2} .

Few-shot Learning Data Splits We also follow the few-shot evaluation protocol used by Marasovic et al. (2022). We use their 60 train-validation splits to run our experiments. Each experiment is run with 48 examples in the training set and 350 examples in the validation set. Note that, depending on the dataset, the number of training examples per label varies: e-SNLI has 16 examples for each label, ECQA 48, SBIC 24, and omVE 24, resulting in 48 training examples for all datasets.

Training Procedure Following Marasovic et al. (2022), we fine-tune T5 (Raffel et al., 2020) and UNIFIEDQA (Khashabi et al., 2020). Depending on the setup, the gradients are activated for specific parameters (SPARSEFIT) or the entire model (baseline). We report our experimental results for the baseline, and observe a consistent behavior with the one reported by Marasovic et al. (2022). Additionally, to compare with other PEFT baselines, we

adapted LoRA (Hu et al., 2022), AdaLoRA (Zhang et al., 2023) and (IA)³ (Liu et al., 2022). We use the PEFT implementation developed by Hugging Face (Mangrulkar et al., 2022). In our implementation of (IA)³, we deviate slightly from its original implementation since we learn scaling vectors for all layers in the model instead of learning them only for the attention modules. This resulted in a significant performance increase. Also, this implementation approach means we are fine-tuning a maximal number of parameters with (IA)³. For the SPARSEFIT configurations, we fine-tune each component (or pair) for 25 epochs with a batch size of 4 samples. Similarly to Marasovic et al. (2022), we use the AdamW optimizer (Loshchilov and Hutter, 2019) with a fixed learning rate of 0.00003. Conditional text generation is used to do the inference on the validation set. Training and evaluation were run on an NVIDIA P100, and took 23.2 min, on average.

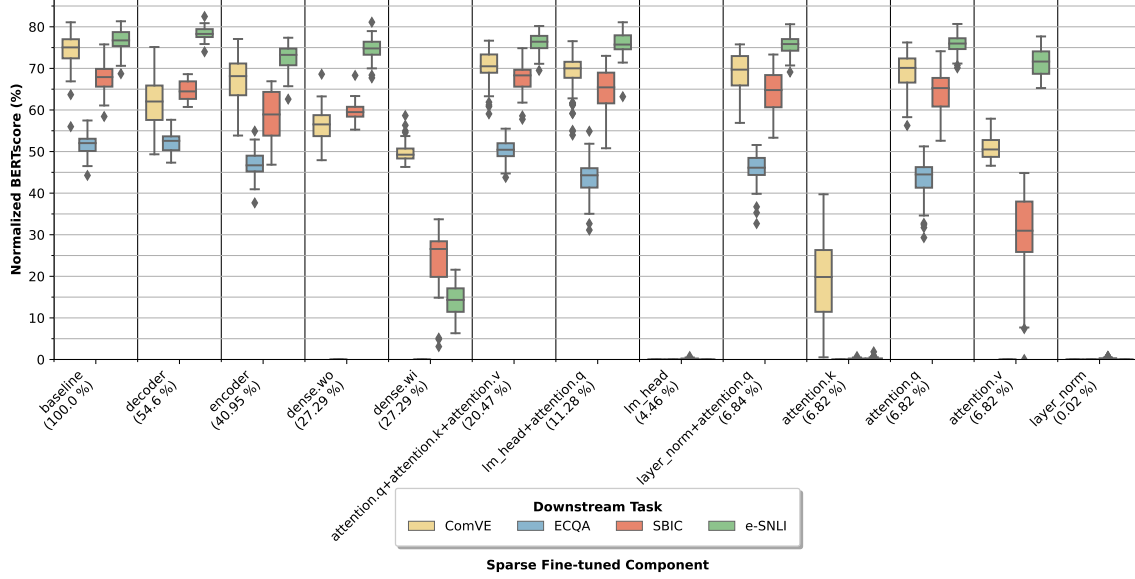


Figure 1: Distribution of the **normalized BERTScore** for different SPARSEFIT settings of sparse fine-tuning for T5-large. The percentage of fine-tuned parameters is shown between brackets.

Automatic Evaluation The evaluation considers the task accuracy and the quality of the generated NLEs. To automatically evaluate the quality of the NLEs, we follow Marasovic et al. (2022) and use the BERTScore (Zhang et al., 2019), which was shown by Kayser et al. (2021) to correlate best with human evaluation in NLEs. As in Marasovic et al. (2022), we compute a **normalized BERTScore** that assigns a zero score to empty NLEs, or NLEs of incorrectly predicted samples (since one would not expect, nor want, an NLE to be plausible if the prediction was wrong). We report the averages and standard deviations of the accuracy and the normalized BERTScore over the 60 train-validation splits for each fine-tuning configuration.

Human Evaluation In addition to the normalized BERTScore, we perform a smaller-scale human evaluation to assess the quality of NLEs for the best-performing SPARSEFIT configurations. We use the NLEs associated with the first 30 correctly predicted samples (balanced to the number of classes) in each validation set for human evaluation. Our human evaluation framework follows those of Kayser et al. (2021); Marasovic et al. (2022). For the NLE quality assessment, each annotator is asked to answer the question: “Does the explanation justify the answer?” and select one of four possible answers: *yes*, *weak yes*, *weak no*, or *no*.

Moreover, we also ask the annotators to identify the main shortcomings, if any, of the generated NLEs. The possible shortcomings categories are

(1) nonsensical, (2) contradictory, (3) lack of explanation, (4) incomplete explanation, (5) input repetition, (6) hallucination, (7) extra words at the end, (8) true but uncorrelated, (9) inaccurate, and (10) one word. An author and a third-party annotator performed independent annotations of the whole set of NLEs chosen to be evaluated (600 examples in total). As in Kayser et al. (2021), we compute a numerical value for the quality of the NLEs by mapping the four answers as follows: *yes* $\rightarrow$ 1, *weak yes* $\rightarrow$ $\frac{2}{3}$, *weak no* $\rightarrow$ $\frac{1}{3}$, and *no* $\rightarrow$ 0; and averaging over all annotations per model.

4.1 Results

To evaluate SPARSEFIT, we compute the task accuracy and the quality of the generated NLEs. Given that there are 62 possible configurations (single layers plus pairs of layers), for space reasons, the following shows the best configurations based on the model’s generalization properties. The results for all configurations are shown in Appendix C.

Task Performance We present in Table 1 the accuracy performance for selected SPARSEFIT settings for T5-large. As can be observed in Table 1, some SPARSEFIT configurations with very few fine-tuned parameters can produce significantly better results than the baseline (i.e., full fine-tuning). For instance, fine-tuning the *Normalization Layer (LayerNorm)* (0.02% of the model’s parameters) achieves better task performance than the baseline for two out of four datasets (e-SNLI

	FLOPS		ComVE	ECQA	SBIC	e-SNLI	Avg
SPARSEFIT (Att.Q+LN) (6.84%)	2.37e14	Acc. nBERTs	74.86 ± 5.27 69.02 ± 4.83	55.81 ± 3.12 45.88 ± 3.72	66.99 ± 4.4 64.29 ± 4.7	82.62 ± 2.73 75.63 ± 2.51	70.07 ± 3.88 63.7 ± 3.94
AdaLoRA (4.48%)	2.87e14	Acc. nBERTs	19.43 ± 1.47 ∇ 16.26 ± 1.23 ∇	59.40 ± 2.28 ∇ 48.30 ± 1.85 ∇	0.18 ± 0.20 ∇ 0.15 ± 0.16 ∇	86.66 ± 1.79 ∇ 72.19 ± 1.49 ∇	41.42 ± 1.44 34.23 ± 1.18
AdaLoRA (1.15%)	1.48e15	Acc. nBERTs	69.66 ± 3.47 ∇ 64.06 ± 3.19 ∇	46.60 ± 4.02 ∇ 41.22 ± 3.65 ∇	61.80 ± 2.74 ∇ 58.91 ± 2.86 ∇	84.50 ± 1.95 ∇ 77.43 ± 1.79 ∇	65.64 ± 3.05 60.41 ± 2.87
LoRA (Att.QV, Rank=128) (4.86%)	2.88e14	Acc. nBERTs	67.77 ± 3.73 ∇ 61.36 ± 3.41 ∇	43.51 ± 3.57 ∇ 0.33 ± 0.41 ∇	63.57 ± 3.16 ∇ 61.06 ± 3.29 ∇	84.26 ± 1.92 ∇ 76.49 ± 1.75 ∇	64.78 ± 3.1 49.81 ± 2.22
LoRA (Att.KQVO, Rank=4) (0.32%)	2.75e14	Acc. nBERTs	68.96 ± 3.68 ∇ 63.48 ± 3.39 ∇	39.04 ± 4.06 ∇ 33.52 ± 3.76 ∇	62.66 ± 3.46 ∇ 59.80 ± 3.66 ∇	84.05 ± 1.81 ∇ 77.04 ± 1.66 ∇	63.68 ± 3.25 58.46 ± 3.12
(IA) ³ (0.07%)	2.74e14	Acc. nBERTs	58.53 ± 2.32 ∇ 53.87 ± 2.15 ∇	59.14 ± 2.36 ∇ 48.08 ± 1.92 ∇	48.08 ± 0.81 ∇ 48.06 ± 0.80 ∇	86.64 ± 1.85 ∇ 72.18 ± 1.54 ∇	63.10 ± 1.84 55.55 ± 1.60

Table 2: Performance comparison between SPARSEFIT and other PEFT strategies. We report the average and the standard deviation over the 60 few-shot train-validation splits for the **accuracy** metric and the normalized BERTScore (**nBERTs**). We show in **bold** the setting with the highest metric for each dataset. Significance testing was assessed via mean t-test in comparison with SPARSEFIT: ∇ represents a p-value lower than 10^{-2} .

and ECQA). Furthermore, we consistently see that if two SPARSEFIT configurations achieve good generalization results, combining them by jointly fine-tuning both components produces significantly better results than each configuration in isolation. We show in Figure 10 the spread of the task performance for SPARSEFIT configurations. Results for T5-base and T5-3b are shown in Appendix C. We found that the task accuracy is consistently higher for the largest LMs for all datasets, but the gap between T5-large and T5-3b is small ($<7\%$) compared with the increase in trainable parameters ($\sim 5\times$).

NLE Quality Recall that the LM is fine-tuned to conditionally generate a text in the form of “[label] because [explanation]”. Figure 1 shows the normalized BERTScore for selected SPARSEFIT settings as a proxy to evaluate how good the NLEs generated after the explanation token (i.e. “because”) is. For all the box plots, the x -axis shows the SPARSEFIT configurations, with the percentage of fine-tuned parameters between brackets. Overall, it can be observed that SPARSEFIT settings with few trainable parameters ($<10\%$), such as the *Self-attention Query* (Att.Q), *LM Head + Attention Query* (Att.Q+LMhead), and *Layer Norm + Attention Query* (Att.Q+LN), are competitive against the baseline for all datasets. Moreover, we can see that the best quality of NLEs is achieved for SPARSEFIT combinations of two or more types of components (e.g., Att.Q). Remarkably, fine-tuning the decoder block ($\sim 54\%$ params) achieves better performance than the entire fine-tuning for two out

of four datasets (e-SNLI and ECQA). The performance gap between most of the SPARSEFIT configurations and the baseline does not exceed 15% for all the datasets, even for very sparse fine-tuning strategies.

Unexpectedly, many SPARSEFIT configurations with high task accuracy (e.g., *LayerNorm*) have a normalized BERTScore close or equal to zero. This happens because either the conditional generation ends the sentence after the generated label token or the explanation token (i.e., “because”) is not successfully generated. We investigate more about this behavior in Section 4.2. We summarize our results on NLEs quality for T5-large in Table 1. Results for other T5 model sizes (i.e. T5-base, T5-large and T5-3b) are shown in Appendix C.1. We found that the normalized BERTScore consistently increases with the size of the LM. Remarkably, the best SPARSEFIT configurations for T5-large also achieve the best performance when fine-tuning T5-base, but they are slightly different for T5-3b.

Other PEFT Baselines To compare SPARSEFIT with other PEFT baselines, we also evaluated LoRA (Hu et al., 2022), AdaLoRA (Zhang et al., 2023) and (IA)³ (Liu et al., 2022) for NLEs. Table 2 shows the downstream performance and the NLEs quality for different PEFT strategies on T5-large. It can be seen that, on average, SPARSEFIT outperforms the other PEFT strategies. While these PEFT methods tune less than 20% of the 50.45 million parameters updated by SPARSEFIT, the quality of NLEs is considerably better for

		ComVE	ECQA	SBIC	e-SNLI	Avg
Baseline - Full Fine-tuning (100%)	Acc.	63.71 $\pm$ 9.14	11.14 $\pm$ 3.14	63.86 $\pm$ 1.86	34.91 $\pm$ 0.43	43.41 $\pm$ 3.64
	nBERTs	55.93 $\pm$ 9.16	9.46 $\pm$ 2.79	57.42 $\pm$ 1.32	28.62 $\pm$ 0.84	37.86 $\pm$ 3.53
SPARSEFIT (Att.Q+LN) (7.97%)	Acc.	68.03 $\pm$ 8.24	24.53 $\pm$ 3.34	57.90 $\pm$ 1.70	40.10 $\pm$ 4.03	47.64 $\pm$ 4.33
	nBERTs	58.67 $\pm$ 7.20	20.60 $\pm$ 2.83	50.41 $\pm$ 2.72	34.32 $\pm$ 3.50	41.00 $\pm$ 4.06
AdaLoRA (0.30%)	Acc.	64.23 $\pm$ 2.86	13.04 $\pm$ 2.09	57.29 $\pm$ 1.86	38.15 $\pm$ 4.22	43.18 $\pm$ 2.76
	nBERTs	56.16 $\pm$ 2.77	11.13 $\pm$ 1.79	50.63 $\pm$ 0.33	33.48 $\pm$ 3.71	37.85 $\pm$ 2.15

Table 3: Performance comparison between SPARSEFIT and other PEFT strategies for **Llama 2-7B**. We report the average and the standard deviation over the 60 few-shot splits for the **accuracy** metric and the normalized BERTScore (**nBERTs**). We show in **bold** the setting with the highest metric for each dataset. Significance testing was assessed via mean t-test in comparison with SPARSEFIT: ∇ represents a p-value lower than 10^{-2} .

Human Evaluation					
	e-SNLI	ECQA	SBIC	ComVE	Avg
Full Fine-tuning	29.63 (0.43)	41.92 (0.23)	54.44 $\pm$ 0.7	21.67 (0.22)	36.91
SPARSEFIT Att.Q	17.28 0.38	35.35 (0.31)	61.11 (0.77)	28.89 (0.35)	35.66
AdaLora 1.15%	23.33 (0.34)	34.44 (0.26)	61.11 (0.69)	23.34 (0.25)	35.55
SPARSEFIT Att.Q+LN	38.27 (0.34)	31.31 (0.26)	58.89 (0.69)	40.00 (0.25)	42.12

Table 4: Average scores given by human annotators for the best performing SPARSEFIT and other PEFT baselines of T5-large. The best results are in **bold**. In brackets, we show the inter-annotator agreement score.

SPARSEFIT for two out of four datasets. Notice that in Table 2 SPARSEFIT has the lowest FLOPS. We hypothesise that this happens since SPARSEFIT neither introduces additional model parameters nor increases the model’s architectural complexity. In Table 8 in Appendix C.3, we show further results for LoRA trained on a bigger range of the number of parameters.

Larger Language Models To evaluate the performance of SPARSEFIT in both larger language models and different architectures we perform a set of experiments applying SPARSEFIT to Llama 2-7B. Notice that SPARSEFIT approach applies to any architecture (not only the T5 encoder-decoder) since it focuses on identifying the optimal layer to fine-tune, regardless of the underlying model’s structure. In this regard, we conducted experiments on a larger decoder-only model (i.e. Llama 2-7B). Table 3 shows the average predictive accuracy and the NLEs quality for the the best SPARSEFIT strategy (Att.Q+LN) and the best performing PEFT baseline (AdaLora) for Llama 2-7B. Overall, it can be observed that SPARSEFIT outperforms the other PEFT strategy for all datasets.

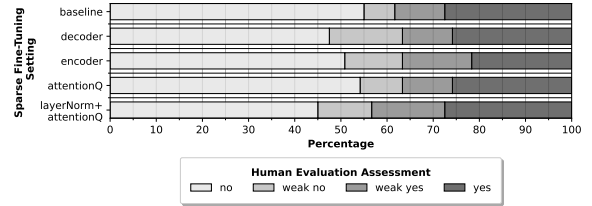


Figure 2: Illustration of plausibility score given by human annotators to the quality of the NLEs generated by different SPARSEFIT configurations. The annotators were asked to answer the question: “Does the explanation justify the answer?”

Particularly, the best SPARSEFIT have on average roughly 5% better NLE quality than the other PEFT.

Human Evaluation We show in Table 4 the distributions of the scores given by the human annotators for the quality of the generated NLEs for the best SPARSEFIT strategies and the best performing PEFT baseline (AdaLora). We compute the inter-annotator agreement score using the *Cohen’s* κ metric (Cohen, 1960). Overall, we found that the quality of the NLEs generated after applying SPARSEFIT is much higher than those of the baseline and AdaLoRA for 2 out of 4 tasks. For the other two tasks, AdaLoRA produces better NLEs by a very smaller margin. On average, the NLEs of SPARSEFIT are roughly 8% better than NLEs of AdaLoRA and 6% better than full fine-tuning NLEs. However, the human evaluation shows that the generated NLEs are often insufficient to explain the predictions accurately. We show in Figure 2 the distributions of the plausability score given by the human annotators for the quality of the generated NLEs for the best SPARSEFIT strategies. It can be observed that roughly half of the NLEs do not justify the answer, no matter what fine-tuning strategy is used. Similarly, the proportion of NLEs

	Human Evaluation				
	e-SNLI	ECQA	SBIC	ComVE	Avg
Full Fine-tuning	17.78	38.89	47.78	40.00	36.11
SPARSEFIT Att.Q+LN	41.11	73.33	68.89	70.00	63.33
AdaLora	50.00	52.22	71.11	55.56	57.22

Table 5: Average scores given by human annotators for the best performing SPARSEFIT and other PEFT baselines of Llama-2-7B. The best results are in **bold**.

that fully justify the prediction is close to 25% regardless of the SPARSEFIT setting. We detail the shortcomings and limitations of generated NLEs in Section 4.2. Finally, we show in Table 5 the human evaluation results for the best SPARSEFIT configuration and other PEFT baselines when applied to Llama2-7B. It can be seen that on average the NLEs of SPARSEFIT are roughly 6% better than NLEs of AdaLoRA and 21% better than full fine-tuning NLEs of Llama2-7B.

4.2 Discussion

Analysis of the Generated NLEs In Figure 4, we show a collection of examples of the generated NLEs by the baseline and the best performing SPARSEFIT configurations. As in previous works (Camburu et al., 2018; Kayser et al., 2021; Marasovic et al., 2022), we only show examples where the label was correctly predicted by the model since we do not expect a model that predicted a wrong label to generate a correct explanation. We present in Appendix B a more extensive list of generated NLEs with SPARSEFIT.

NLE Shortcomings Figure 3 depicts the histogram of frequencies of the shortcomings for the baseline and the best-performing SPARSEFIT strategies. It can be observed that the most common shortcomings are *Lack of explanation*, *Nonsensical*, and *Incomplete explanation*. For the best SPARSEFIT configuration (i.e. Att.Q+LN), the *Incomplete explanation* is the reason with the most occurrences. We show a breakdown of the shortcomings for each dataset in Appendix C.4.

Inter-Annotator Agreement As shown in Table 4, the agreement between annotators is moderately low for the set of evaluated NLEs. More precisely, the annotators gave different scores to 181 out of 600 NLEs. The dataset with the most significant difference is ECQA, with 63 differences,

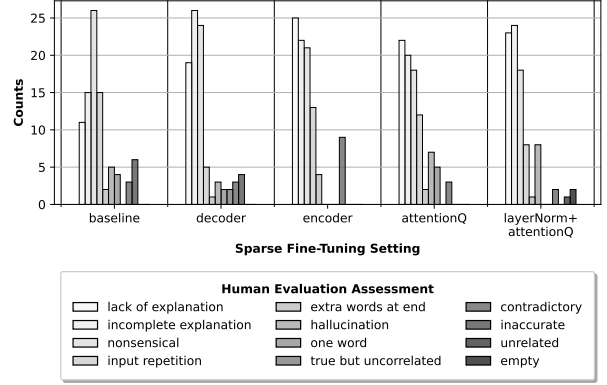


Figure 3: Histogram of the shortcomings of the generated NLEs for the baseline and the performing SPARSEFIT configurations aggregated for all the datasets.

while the SBIC dataset is the most uniform, with 17 differences. The variation between annotators can result from three potential perceptual reasons (Bourke, 2014; Niño, 2009). The first reason is the *perceptual disagreement*, which states that annotators could not objectively identify the difference between two adjacent answers (i.e. *Weak Yes* vs *Weak No*, or *Yes* vs. *Weak Yes*). The second reason is *positionality disagreement* (Bourke, 2014), which could alter how the annotators perceive the outcomes of the algorithm due to their race, gender, and other socioeconomic identity factors. This is particularly crucial for the SBIC dataset, as it contains offensive content. The third reason is the *expectation disagreement*, which may cause an annotator to be more strict on the characteristics that make an explanation complete and accurate. An extensive collection of examples of perceptual disagreement, positionality disagreement, and expectation disagreement samples are in Appendix C.5.

Generation of Empty NLEs As mentioned earlier, some SPARSEFIT configurations (e.g. *LayerNorm*) have high task performance but generate empty NLEs most of the time, particularly for the e-SNLI and ECQA datasets. One possible explanation for the discrepancy between the high task accuracy and the low NLE quality is that generating NLEs is an intrinsically more complex problem than solving the downstream tasks, where the former may require fine-tuning more significant portions of the model parameters. Another explanation can be found by analyzing the pre-training tasks of the PLM and observing that, in the pre-training stage, T5 was trained on the MNLI dataset (Williams et al., 2018), which is composed

Premise Hypothesis Label Explanation	2 woman walking down the street next to graffiti The 2 woman are running Contradiction You can't be running, if you are walking. Walking is different from running.
Baseline Decoder Att.Q Att.Q + Head Att.Q + LN	Generated Explanation walking and running are different actions walking and running are different verbs the 2 woman are walking not running walking and running are not the same walking is not running

Question Answer Choices Answer Explanation	The teacher doesn't tolerate noise during a test in their what? A) Theatre B) Bowling C) Alley D) Factory E) Classroom Classroom Test is done in classroom. A teacher when in classroom does not tolerate noise.
Baseline Decoder Att.Q Att.Q + Head Att.Q + LN	Generated Explanation noise is harmful for students. classroom is a place where students can take test. classroom is a place where students are studying. there is no noise in classroom. classroom is a place where people can learn. noise is a problem in classroom. noise is harmful to students classroom is a place where people can learn. noise is a distracting thing.

Post Label Explanation	@cmencarini: I dont trust girls ; cant trust these hoes Offensive This post implies that women are not honest. This post implies that women are not trustworthy. This post implies that women are all women are hoes.
Baseline Decoder Att.Q Att.Q + Head Att.Q + LN	Generated Explanation this post implies that women aren't important this post implies that women are prostitutes this post implies that women aren't trustworthy this post implies that women aren't trustworthy this post implies that women are dirty

Choice 1 Choice 2 Label Explanation	A Latino can communicate with someone from Spain French and Latinos can communicate easy Choice 2 French can't communicate with Latinos because they don't speak the same language. French and Latinos belongs to two different countries with two different languages.
Baseline Decoder Att.Q Att.Q + Head Att.Q + LN	Generated Explanation french and latinos are two different languages french and latinos are two different people and don't speak the same language french is not a common language in latinamerica french and latinos cannot communicate easily. french and latinos cannot communicate easily

Figure 4: Examples of generated NLEs for e-SNLI (Green), ECQA (Blue), SBIC (Red), and ComVE (Yellow).

of NLI instances without NLEs. The T5 weights were then pre-trained on MNLI by casting the NLI task as a sequence transduction problem, where the input is a hypothesis-premise pair, and the output is the label. When only a small subset of parameters is updated (e.g., *LayerNorm*(0.02%)), the model elicits its original behavior and predicts the label without generating the NLE. Similar reasoning may be concluded for ECQA since UNIFIEDQA was pre-trained on CommonsenseQA (Talmor et al., 2019), which is composed of samples with only the

answer for the multiple-choice question.

5 Summary

We introduced SPARSEFIT, a strategy that combines sparse fine-tuning with prompt-based learning to train NLE models in a few-shot setup. SPARSEFIT shows consistently competitive performance while only updating a minimal subset of parameters (i.e. the *Self-attention Query + Layer Normalization*, having $\sim 6.8\%$ of the model parameters). We found that the sparse fine-tuning of T5-large consistently achieves better performance than fine-tuning T5-base and is slightly worse ($< 5\%$) than T5-3b, no matter the SPARSEFIT strategy. Moreover, the top three best performers for T5-base are achieved by the same set of SPARSEFIT configurations found for T5-large. Compared to other PEFT techniques, SPARSEFIT produces better average predictive accuracy and NLE quality. We aim for SPARSEFIT to inspire the community to investigate sparse fine-tuning at different model components.

Limitations

Although generating natural language explanations is a fervid research area, there is still no guarantee that such explanations accurately reflect how the model works internally (Wiegrefe et al., 2021; Camburu et al., 2020). For example, the fact that the generated explanation seems reasonable does not mean that the model does not rely on protected attributes and spurious correlations in the training data to produce its predictions. As such, we still recommend being careful to use self-explanatory models in production, as they can capture potentially harmful biases from the training data, even though these are not mentioned in the explanations.

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A SPARSEFIT Graphical Representation

In this paper, we propose an efficient few-shot prompt-based training regime for models generating both predictions and NLEs on top of the T5 language model. To have a better understanding of the active trainable parameters in each SPARSEFIT configuration, we illustrate in Figure 5 a graphical representation of the T5 architecture with active parameters colored for the *Layer Normalization* sparse fine-tuning. After freezing the rest of the model (gray-colored layers), the percentage of parameters that could potentially be updated in the *Layer Normalization* is 0.02% of the entire model. Considering that the UNIFIEDQA model’s architecture is the same as the one in T5, the interpretation of active parameters holds for UNIFIEDQA.

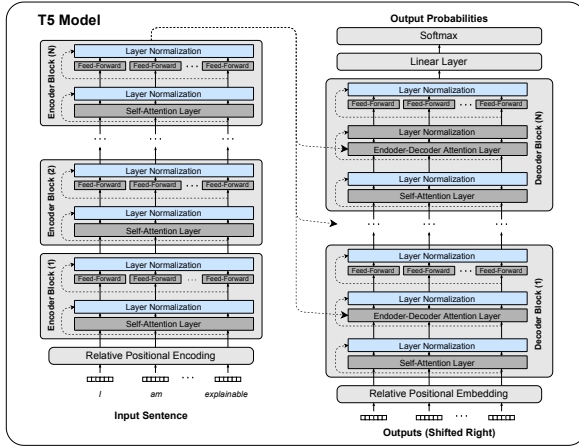


Figure 5: Illustration of the active trainable parameters in T5 when SPARSEFIT is performed over the layer normalization.

B Examples of Generated NLEs

This section shows a collection of examples of the generated NLEs by the baseline and the different sparse fine-tuning strategies considered in our approach. We show four examples for each dataset. Each example contains the generated NLE for the best performing SPARSEFIT configurations. As in previous works (Camburu et al., 2018; Kayser et al., 2021; Marasovic et al., 2022), we only show examples where the label was correctly predicted by the model (since we do not expect a model that predicted a wrong label to generate a correct NLE). Regarding the protocol for choosing the examples shown in this section, we have done a manual inspection of several possible examples, and we have chosen the more informative ones to conclude the strengths and the weaknesses of the generated

NLEs. Notice that, due to the few-shot splits protocol (60 different train-validation splits), a single example could be predicted more than once for a single setup (i.e., the sample is in more than one validation set).

Premise	Two dogs play together on the bed.
Hypothesis	Dogs laying down on the floor, motionless.
Label	Contradiction
Generated Explanation	
Baseline	dogs can't be motionless and play together on the bed.
Decoder	playing together is different than lying down, motionless.
Encoder	dogs laying down on the floor are motionless.
Attention.Q	two dogs playing together on the bed are not laying down, motionless.
LM Head + Att.Q	two dogs playing together on the bed are not motionless.
LayNorm + Att.Q	two dogs playing together on the bed are not laying down, motionless.
Self-Attention	two dogs playing together on the bed are not motionless.

Premise	Man walking barefoot on the beach on what seems to be a very cloudy day
Hypothesis	It is going to rain soon.g
Label	Neutral
Generated Explanation	
Baseline	it does not look like it is going to rain soon
Decoder	not all cloudy means it is raining
Encoder	it does not mean it is going to rain soon
Attention.Q	not all clouds are bad
LM Head + Att.Q	not all cloudy days are going to rain soon
LayNorm + Att.Q	it does not mean it is going to rain soon
Self-Attention	not all cloudy days are going to rain soon

Premise	Number 13 kicks a soccer ball towards the goal during children's soccer game
Hypothesis	A player trying to score in a soccer game
Label	Entailment
Generated Explanation	
Baseline	number 13 is a player
Decoder	number 13 is a player trying to score
Encoder	number 13 is a player trying to score
Attention.Q	number 13 is a player
LM Head + Att.Q	number 13 is a player trying to score
LayNorm + Att.Q	number 13 is a player
Self-Attention	kicking a soccer ball towards the goal is similar to trying to score

Figure 6: Examples of generated NLEs for e-SNLI.

Question Answer Choices Answer Baseline Decoder Encoder Attention.Q LM Head + Att.Q LayNorm + Att.Q Self-Attention	What does a person who likes school also do? A) Fever B) Play dumb C) Read text D) Catch cold E) Enjoy learning Enjoy learning Generated Explanation A school is a place where one can learn. it is a place where one can get knowledge and skills a person who likes school also does. a person who likes school also does enjoy learning is also a part of life of a person. a person who likes school also does enjoy learning school is a place where people learn. a person who likes school also likes to learn. school is a place where people learn. a person who likes school also does things like studying. school is a place where people learn. a person who likes school also does things like studying. a person who likes school also does things like studying.
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Question Answer Choices Answer Baseline Decoder Encoder Attention.Q LM Head + Att.Q LayNorm + Att.Q Self-Attention	When someone is watching a film that is a documentary, what are they hoping to gain? A) Like B) Insight C) Fear D) Being entertained E) Awareness Insight Generated Explanation insight is the ability to gain knowledge through observation. when someone watches a documentary, they are hoping to gain insight. insight is the ability to gain knowledge through observation. people often seek out information through watching a documentary. insight is the ability to gain knowledge through observation. insight is a kind of knowledge that is gained through observation. insight is a quality that is gained when someone is watching a documentary. insight is gaining knowledge. people are hoping to gain knowledge by watching a documentary. insight is the ability to gain knowledge. when someone is watching a documentary, they are hoping to gain knowledge.
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Question Answer Choices Answer Baseline Decoder Encoder Attention.Q LM Head + Att.Q LayNorm + Att.Q Self-Attention	When you go on vacation what do you have to do with things you want to take with you? A) Pay bills. B) Save money. C) Pack. D) Sell. E) Have money Pack Generated Explanation packing means to take something with you. if you want to take something with you, you have to pack it. packing means packing to take with you when you go on vacation. if you want to take things with you, you have to pack them. packing is the process of packing things you want to take with you. packing is a process of taking things with you. when you go on vacation you have to pack things you want to take with you when you go on vacation you have to pack things you want to take with you. tpacking is a process where you put things together to take with you when you go on vacation. you have to pack things when you go on vacation. packing means taking things with you.
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Post Label Baseline Decoder Encoder Attention.Q LM Head + Att.Q LayNorm + Att.Q Self-Attention	Watching conservacucks slobber all over the Western Wall makes me ill. Offensive Generated Explanation this post implies that southerners are lazy. this post implies that conservacucks are slobbering this post implies that conservacucks are disgusting this post implies that conservacucks are slobbering all over the western wall. tthis post implies that conservacucks are slobbering all over the western wall. this post makes fun of conservacucks this post implies that conservacucks are dirty.
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Post Label Baseline Decoder Encoder Attention.Q LM Head + Att.Q LayNorm + Att.Q Self-Attention	Whats the difference between a training gun and a police officer? A training gun shoots blanks, an officer shoots blacks Offensive Generated Explanation this post implies that black people are criminals this post implies that blacks are the only color that will be killed this post implies that blacks are the victims of police brutality this post implies that blacks are more dangerous this post implies that blacks are more dangerous this post implies that black people are criminals this post implies that blacks are inferior
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Post Label Baseline Decoder Encoder Attention.Q LM Head + Att.Q LayNorm + Att.Q Self-Attention	What's the worst than a nerd being Basketball Player A Nigga Becoming a Doctor Offensive Generated Explanation this post implies that black folks are less than white this post trivializes harm to victims this post implies that blacks are worse than nerd this post implies that blacks are better than nerds this post implies that blacks are inferior this post implies that blacks are inferior this post implies that blacks are the only people who shoot training guns
---	---

Figure 8: Examples of generated NLEs for SBIC.

Figure 7: Examples of generated NLEs for ECQA.

Choice 1	He used a towel to dry off after his shower.
Choice 2	He used an apple to dry off after his shower.
Label	Choice 2
Generated Explanation	
Baseline	apple is a fruit, so it cannot dry off.
Decoder	apple is a fruit and does not dry off
Encoder	an apple is not a very absorbent material
Attention.Q	an apple doesn't dry off as well as a towel
LM Head + Att.Q	an apple is a fruit and it doesn't dry off.
LayNorm + Att.Q	an apple is a fruit and it's not a towel, so it's not drying off as well.
Self-Attention	apples are a fruit, and they don't dry well on their own.

Choice 1	He walked his fish
Choice 2	He walked his dog
Label	Choice 1
Generated Explanation	
Baseline	fish have no legs
Decoder	fish cannot walk on water
Encoder	fish cannot walk
Attention.Q	fish are not able to walk
LM Head + Att.Q	fish don't walk
LayNorm + Att.Q	fish cannot walk
Self-Attention	fish cannot be walked

Choice 1	A ball is square and can roll
Choice 2	A ball is round and can roll
Label	Choice 1
Generated Explanation	
Baseline	a square ball cannot roll
Decoder	square balls are not rollable
Encoder	a ball is square cannot roll
Attention.Q	balls are round and can roll
LM Head + Att.Q	a ball is round and can roll.
LayNorm + Att.Q	a square ball cannot roll
Self-Attention	a ball can roll only in a round shape

Figure 9: Examples of generated NLEs for ComVE.

C SPARSEFIT Full Results

This section shows the results in terms of task accuracy, and NLEs quality all configurations of SPARSEFIT and for different model sizes (i.e. T5-base, T5-large and T5-3b). For each metric, we also break down the results by dataset.

C.1 Task Performance

Figure 10 depicts the distribution of the accuracy score for SPARSEFIT configurations trained on top of T5-large. It can be observed that several SPARSEFIT configurations exhibit similar performance as the baseline, particularly for ECQA and E-SNLI. The SPARSEFIT configurations with the best task performance are *Decoder*, *Self-Attention KQV*, *Self-attention Query*, and *Layer Normalization*. Remarkably, the SPARSEFIT configurations do not show a higher variance than the baseline across the 60 train-validation splits (inter-quartile range). Figure 11 depicts the distribution of the accuracy score for SPARSEFIT configurations trained on top of T5-3b. It can be observed that all SPARSEFIT configurations outperform the base-

line. However, the best performance for T5-3b is achieved by the sparse fine-tuning of the *Self-attention Value Layer*. The results for T5-base can be observed in the breakdown done for each dataset.

Figure 12 depicts the box plot with the distribution of the accuracy scores on e-SNLI for the 60 train-validation splits for different SPARSEFIT configurations and the two pre-trained LM sizes. Overall, for e-SNLI, the task performance increases with the size of the model for most of the sparse fine-tuning configurations. Moreover, the interquartile range is considerably smaller when the model size increases (i.e., T5-large scores are less spread than the ones for T5-base). The highest median score was achieved by the fine-tuning of the *Layer Normalization* in T5-large, followed very closely by the fine-tuning of the *LM head* and the *Decoder* in T5-large. The combination of components (i.e. Layer Norm + Self-attention Query) performed very closely to the best-performing settings.

For the ECQA dataset, Figure 13 shows the box plot with the accuracy scores for different SPARSEFIT setups. It can be observed that the performance of the larger LM (i.e., T5-large) is consistently better than T5-base. Overall, the accuracy is fairly similar for all the SPARSEFIT configurations for a given LM size, with an average of 58% and 42% for T5-large and T5-base, respectively. Note that the random guess accuracy is 20% for the ECQA dataset, since there are 5 possible answer choices. The highest accuracy was achieved by the fine-tuning of the *Decoder* in T5-large, followed very closely by the fine-tuning of the *Layer Normalization* and *LM Head*. The combination of components achieves a slightly lower performance than single components for the task prediction. Surprisingly, for ECQA, the variability for a given combination of configuration-model (i.e. each box) is higher for T5-large than for T5-base. Moreover, the fine-tuning of the *Encoder* for T5-base gives worse results in comparison with all the other configurations. Besides the setting where only the *Encoder* is fine-tuned for T5-base, the highest observed range in ECQA is roughly 14%.

For the SBIC dataset, Figure 14 depicts the box-plot with the dispersion of accuracy scores for T5-base and T5-large. Recall that for the SBIC dataset, we fine-tune the UnifiedQA variant of T5. In general, it can be seen that

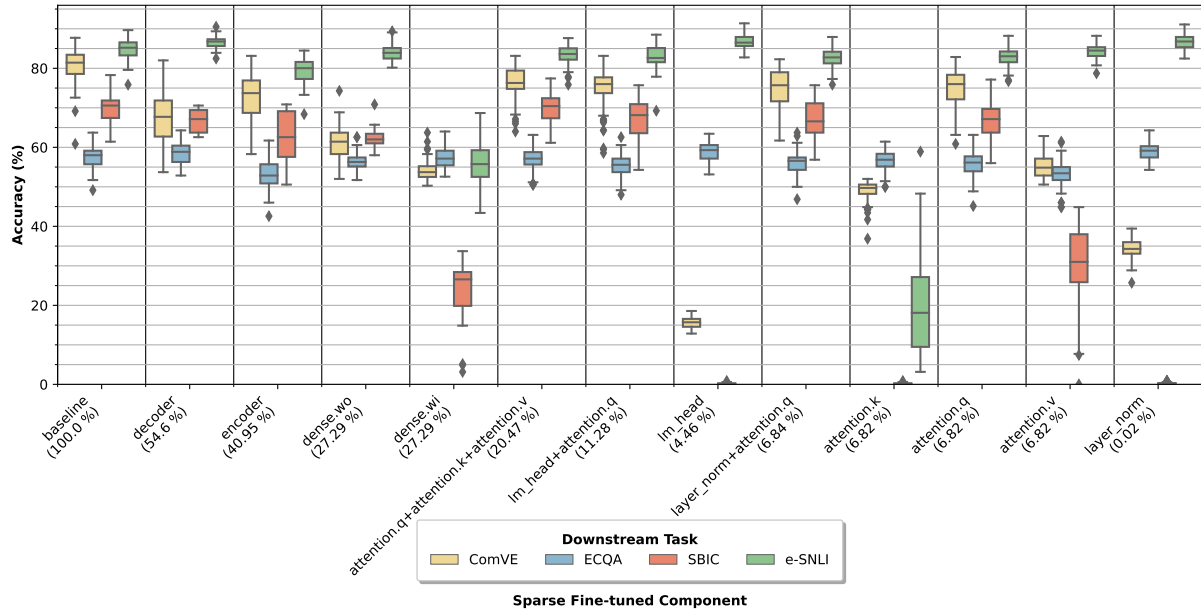


Figure 10: Distribution of the **accuracy** scores for different SPARSEFIT configurations for T5-large. The percentage of parameters fine-tuned for each configuration is shown between brackets.

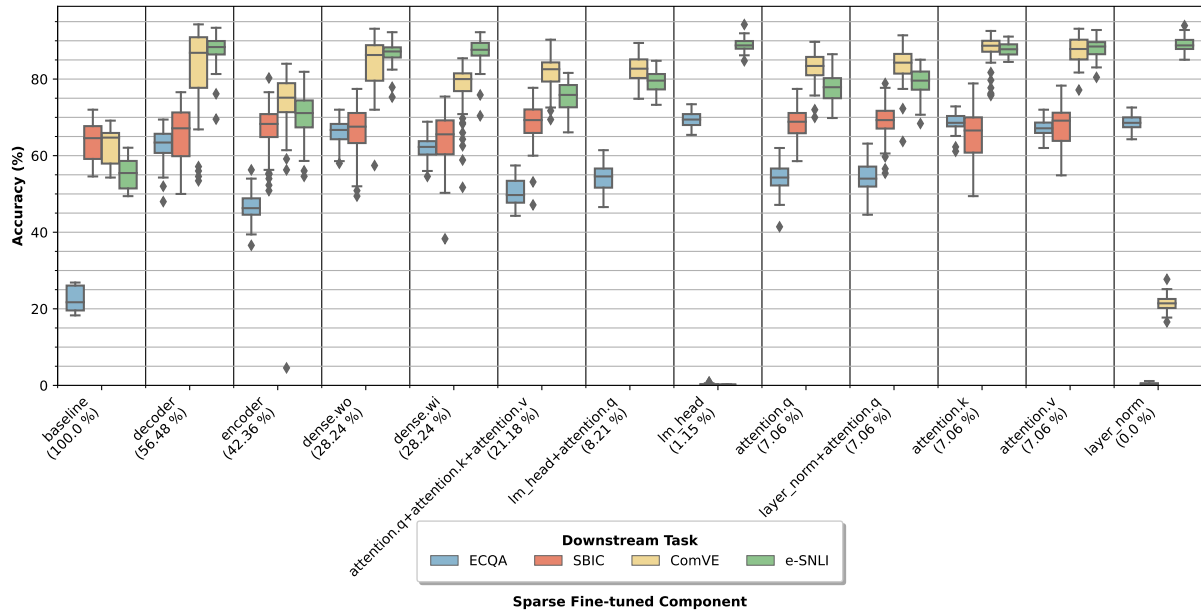


Figure 11: Distribution of the **accuracy** scores for different SPARSEFIT configurations for T5-3b. The percentage of parameters fine-tuned for each configuration is shown between brackets.

SPARSEFIT		ComVE	ECQA	SBIC	e-SNLI	Avg
LayerNorm + Attention.Q	Acc.	53.22 ± 3.67 ∇	39.35 ± 2.31 ∇	62.11 ± 5.04 ∇	72.63 ± 2.87 ∇	56.83 ± 3.47
T5-base	nBERTs	48.77 ± 3.37 ∇	0.0 ± 0.0 ∇	59.45 ± 5.47 ∇	64.81 ± 3.13 ∇	43.26 ± 2.99
LayerNorm + Attention.Q	Acc.	74.9 ± 5.3 ∇	55.8 ± 3.1 ∇	67.0 ± 4.4 ∇	82.6 ± 2.7 ∇	70.1 ± 3.9
T5-large	nBERTs	69.0 ± 4.8	45.9 ± 3.7 ∇	64.3 ± 4.7	75.6 ± 2.5 ∇	63.7 ± 3.9
LayerNorm + Attention.Q	Acc.	83.27 ± 4.52 ∇	54.13 ± 3.86 ∇	68.87 ± 4.86 ∇	79.16 ± 3.72 ∇	71.36 ± 4.24
T5-3B	nBERTs	75.83 ± 4.14 ∇	48.31 ± 3.46 ∇	65.86 ± 5.07 ∇	71.27 ± 3.44 ∇	65.32 ± 4.03

Table 6: Summary of best performing SPARSEFIT configurations for *LayerNorm + Attention*. We report the average and the standard deviation over the 60 few-shot train-validation splits for the **accuracy** metric and the normalized BERTScore (**nBERTs**). In brackets are the percentages of fine-tuned weights for each SPARSEFIT configuration. We show in **bold** the setting with the highest metric for each dataset, in **blue** the highest performance among SPARSEFIT without considering the number of parameters, and in **green** the best-performing setting after considering the percentage of fine-tuned parameters. The trade-off between parameters and performances was

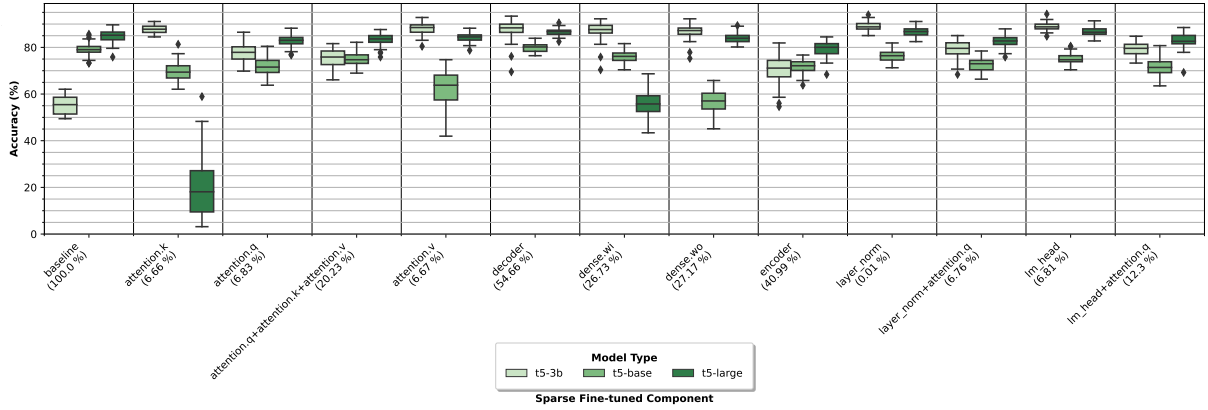


Figure 12: Distribution of the accuracies for different settings of SPARSEFIT for the **e-SNLI** dataset. For each model, the variation represents the overall performance in each of the 60 train-validation splits. The percentage of parameters fine-tuned for each setup is depicted in brackets below the name of each configuration.

the accuracy score surges when the model size is increased; thus, the best accuracy scores for a given sparse fine-tuning setup are found for the T5-large. The best median accuracy performance is achieved by the baseline. However, the difference in the median scores between the best and the second and third best-ranked configurations (i.e. *Self-attention Layer* and *Layer Normalization + Self-attention Query*, respectively) are less than 3%. The maximum variance among scores for the 3 best-performing SPARSEFIT configurations is roughly 15%. Furthermore, it can be observed that for many very sparse fine-tuning configurations, the accuracy score is close to or equal to zero. Even though the performance of a random model is 50%, an accuracy of 0% is feasible in our scenario as the model could generate different words from the ones expected as labels. In this regard, the accuracy scores of zero are a consequence of the fact that, after the conditional generation, the model generates neither “*offensive*” nor “*non-offensive*” for any sample in the validation set. Notice that

this phenomenon is particularly happening when only a small fraction of weights is fine-tuned.

For the ComVE dataset, we show in Figure 15 the accuracy for the 60 different train-validation splits for different SPARSEFIT settings and model sizes. It can be seen that the best-performing setting in terms of accuracy is the baseline for UNIFIEDQA-T5-large. (i.e. *Self-attention Layer* and *Layer Normalization + Self-attention Query* fine-tuning are the second and third best performing, respectively. Overall, the fine-tuning of the *Normalization Layer* leads to the worst task performance. Moreover, it can be observed that the performance increases with the size of the model, thus UNIFIEDQA-T5-large always performs better than UNIFIEDQA-T5-base for all the fine-tuning configurations. The smallest gap in performance between model sizes (UNIFIEDQA-T5-large vs. UNIFIEDQA-T5-base) happens for the fine-tuning of the *Dense Layer*. Conversely, the maximum spread in performance (i.e. the

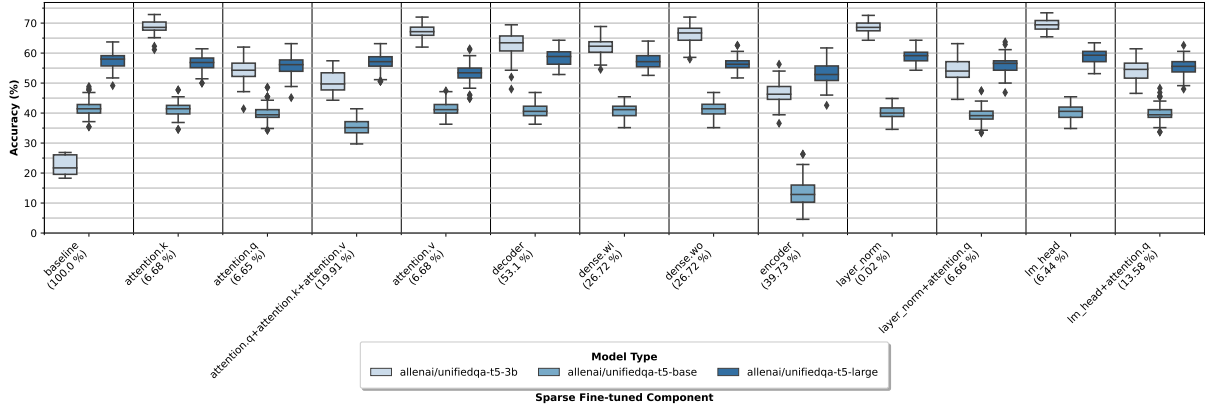


Figure 13: Distribution of the accuracy scores for different SPARSEFIT settings for the **ECQA** dataset. For each model, the variation represents the overall performance in each of the 60 train-validation splits. The percentage of parameters fine-tuned for each setup is depicted in brackets below the name of each configuration. `todoUpdate` plot with t5-3b results

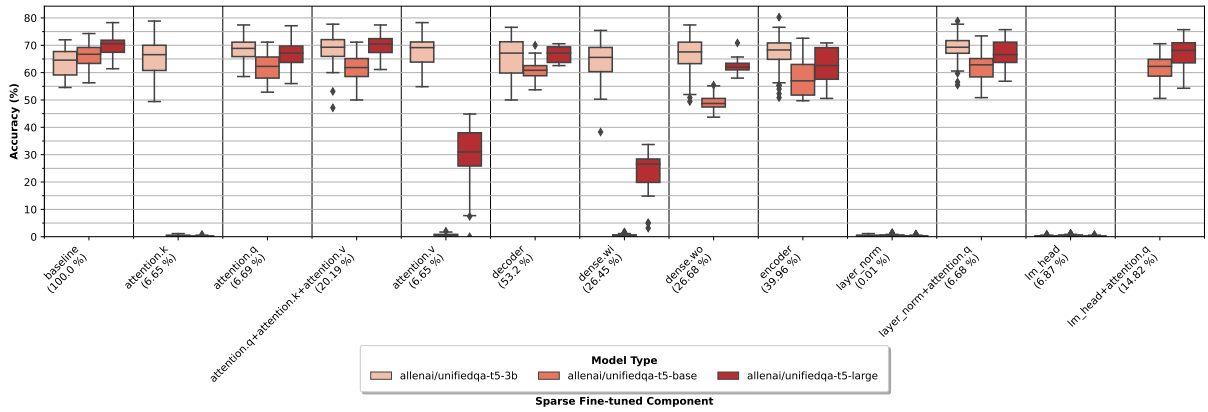


Figure 14: Distribution of the accuracy scores for different settings of SPARSEFIT for the **SBIC** dataset. For each model, the variation represents the overall performance in each of the 60 train-validation splits. The percentage of parameters fine-tuned for each setup is depicted in brackets below the name of each configuration.

difference between the best and the worst split) is around 21% for models trained using the UNIFIEDQA-T5-large architecture.

C.2 Explanation Generation Performance

Figure 1 shows the box-plot with the normalized BERTscores for different SPARSEFIT setups fine-tuned on top of T5-large. In addition to explained in the main text, it can be seen that combinations of components lead to less variance in the score achieved for the 60 train-test splits (see the interquartile range). Furthermore, Table 7 shows the performance summary for the downstream performance and the NLEs quality for T5-3b. It can be observed that the *Attention Value Layer* achieves the best performance on average. We highlight that SPARSEFIT outperforms the baseline (i.e. full fine-tuning) for all datasets.

For e-SNLI, Figure 17 shows the normalized

BERTscore over the 60 few-shot learning splits for different SPARSEFIT configurations. Overall, for every sparse fine-tuning setting, the BERTscore is consistently higher for the largest PLM (i.e. T5-large). However, the gap in performance is smaller for the best-performing sparse fine-tuning configurations. For instance, the difference in the average normalized BERTscore values between T5-large and T5-base for the best performing SPARSEFIT (i.e., *Decoder*) is roughly 5% while for the worst performing configuration is around 68%. The first five best-performing SPARSEFIT configurations for T5-large are *Decoder*, *Baseline*, *Self-attention KVQ*, *Layer Normalization + Self-attention Q*, and *Self-attention Values*. Note that the normalized BERTscore is zero for some sparse fine-tuning configurations (e.g., *Layer Normalization*). This is mostly happening when the sparse fine-tuning is applied to small models (i.e.,

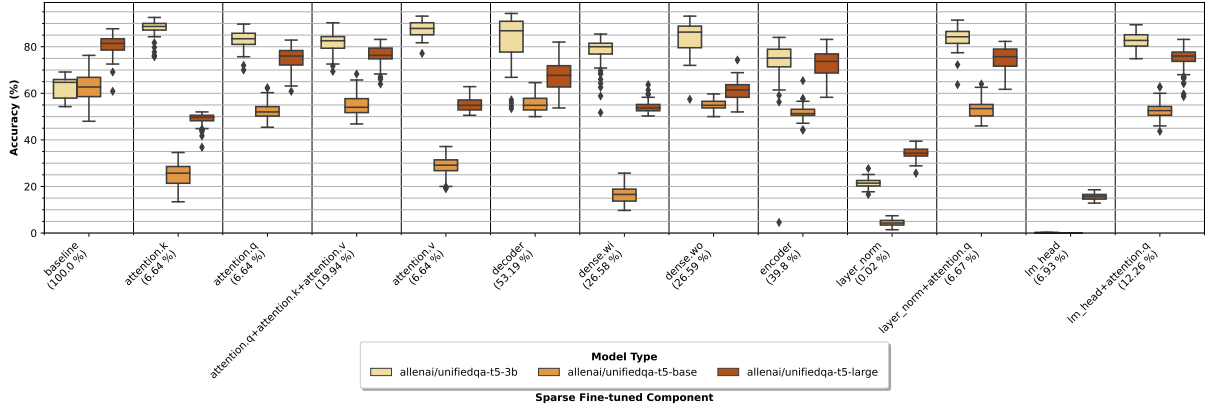


Figure 15: Distribution of the accuracy scores for different settings of SPARSEFIT for the **ComVE** dataset. For each model, the variation represents the overall performance in each of the 60 train-validation splits. The percentage of parameters fine-tuned for each setup is depicted in brackets below the name of each configuration.

T5-base). The fact that the BERTscore is zero for a given configuration for all the samples in a split implies that the generated NLEs are always empty. We explore the reasons behind this phenomenon in Section 4.2

For the ECQA dataset, we show in Figure 18 the spread of the normalized BERTscore for all SPARSEFIT configurations. Without exception, the largest model (T5-large) outperforms the T5-base models for every setting. Remarkably, for ECQA, many sparse fine-tuning configurations lead to the generation of empty explanations. Particularly, only the fine-tuning of the *Baseline*, the *Decoder*, and the *Encoder* are able to consistently generate non-empty explanations no matter the size of the model. Among the configurations that generate non-empty explanations, the best normalized BERTscores are achieved by the *Decoder* sparse fine-tuning, followed by the *Baseline* and *Encoder Blocks* fine-tuning. Note that for all of these configurations, the interquartile range is smaller than 6% regardless of the model size. Moreover, the fine-tuning of *Self-attention Query* achieves competitive results for T5-large but zero BERTscore for T5-base.

Figure 19 shows the normalized BERTscore results for the SBIC dataset. Recall that for the SBIC dataset, we fine-tune the UnifiedQA variant of T5. As expected, the model size contributes to better performance. Consequently, the BERTscore is higher for the T5-large model for every sparse fine-tuning configuration. The best BERTscore median is achieved by the *Baseline* in combination with the UNIFIEDQA-T5-large, with a metric value of $\approx 68\%$. The second and third

best-performing setups are the *Decoder* and the *Encoder*, respectively. Moreover, the fine-tuning of layers such as the *Normalization Layer* or *Self-attention Layer* results in the generation of text that does not contain the explanation token “because”, thus the BERTscore is close to zero for those configurations.

We depict in Figure 20 the variation of the normalized BERTscore metric over the 60 different train-validation splits for the SPARSEFIT configurations. Recall that for ComVE dataset, we fine-tune the UnifiedQA variant of T5. Overall, the BERTscore is substantially higher for T5-large. The best BERTscore for T5-large is obtained by the *Baseline* fine-tuning, with a median score of 75% for the 60 different seeds. Similar behavior can be seen for T5-base, where *Baseline* is also the setting with the best explanations (from the perspective of the automatic metric). The second and third best sparse fine-tuning setups are the *Self-attention Query* and *Baseline*, respectively. Notice that the difference in the median between the *Baseline* and the *Encoder* is around 3%. Moreover, the variance among the different splits for a given sparse fine-tuning setting is on average higher than for the *Baseline*. Remarkably, the sparse fine-tuning over the *Normalization Layer* was the only setting that obtained a zero BERTscore for the ComVE dataset.

C.3 Other PEFT Baselines

In order to make our approach comparable in the number of parameters, we test LoRa (Hu et al., 2022) using higher ranks. Table 8 shows the performance of LoRa for different rank sizes. Notice

SPARSEFIT		ComVE	ECQA	SBIC	e-SNLI	Avg
Baseline (100.00%)	Acc. nBERTs	62.48 ± 6.03 55.55 ± 5.6	22.39 ± 3.61 19.73 ± 3.22	63.55 ± 6.59 61.21 ± 6.79	55.3 ± 4.98 49.25 ± 4.36	50.93 ± 5.3 46.44 ± 4.99
Decoder (54.60%)	Acc. nBERTs	83.67 ± 10.12 ∇ 74.66 ± 9.02 ∇	62.62 ± 4.16 ∇ 55.31 ± 3.65 ∇	65.59 ± 7.51 62.72 ± 7.66	87.48 ± 4.02 ∇ 77.92 ± 3.7 ∇	74.84 ± 6.45 67.65 ± 6.01
Encoder (40.95%)	Acc. nBERTs	73.14 ± 11.24 ∇ 66.7 ± 10.28 ∇	46.23 ± 3.96 ∇ 41.46 ± 3.56 ∇	66.81 ± 6.43 64.45 ± 6.7	70.34 ± 5.79 ∇ 63.79 ± 5.28 ∇	64.13 ± 6.86 59.1 ± 6.46
Dense.wo (27.29%)	Acc. nBERTs	83.91 ± 6.54 ∇ 76.1 ± 6.04 ∇	66.21 ± 3.12 ∇ 59.12 ± 2.76 ∇	66.64 ± 6.46 63.87 ± 6.51	86.85 ± 3.0 ∇ 78.24 ± 2.78 ∇	75.9 ± 4.78 69.33 ± 4.52
Dense.wi (27.29%)	Acc. nBERTs	77.6 ± 6.63 ∇ 70.21 ± 6.04 ∇	62.12 ± 2.75 ∇ 55.12 ± 2.44 ∇	63.99 ± 7.4 61.05 ± 7.43	87.31 ± 3.6 ∇ 78.24 ± 3.28 ∇	72.76 ± 5.1 66.16 ± 4.8
Attention KQV (20.47%)	Acc. nBERTs	81.73 ± 4.14 ∇ 74.27 ± 3.84 ∇	50.24 ± 3.48 ∇ 44.79 ± 3.06 ∇	68.84 ± 5.37 ∇ 66.12 ± 5.46 ∇	75.3 ± 3.78 ∇ 67.67 ± 3.36 ∇	69.03 ± 4.19 63.21 ± 3.93
LM head + Attention.Q (11.28%)	Acc. nBERTs	82.59 ± 3.37 ∇ 75.2 ± 3.0 ∇	54.28 ± 3.57 ∇ 48.42 ± 3.17 ∇	0.0 ± 0.0 0.0 ± 0.0	79.33 ± 2.94 ∇ 71.52 ± 2.74 ∇	72.07 ± 3.29 65.05 ± 2.97
LM head (4.46%)	Acc. nBERTs	0.09 ± 0.13 ∇ 0.0 ± 0.0 ∇	69.43 ± 1.88 ∇ 0.0 ± 0.0 ∇	0.23 ± 0.22 ∇ 0.19 ± 0.18 ∇	89.04 ± 1.63 ∇ 0.0 ± 0.0 ∇	39.7 ± 0.96 0.05 ± 0.04
LayerNorm + Attention.Q (6.84%)	Acc. nBERTs	83.27 ± 4.52 ∇ 75.83 ± 4.14 ∇	54.13 ± 3.86 ∇ 48.31 ± 3.46 ∇	68.87 ± 4.86 ∇ 65.86 ± 5.07 ∇	79.16 ± 3.72 ∇ 71.27 ± 3.44 ∇	71.36 ± 4.24 65.32 ± 4.03
Attention.Q (6.82%)	Acc. nBERTs	83.09 ± 4.15 ∇ 75.65 ± 3.76 ∇	54.39 ± 3.66 ∇ 48.56 ± 3.24 ∇	68.44 ± 4.44 ∇ 65.4 ± 4.68 ∇	77.88 ± 3.66 ∇ 70.23 ± 3.41 ∇	70.95 ± 3.98 64.96 ± 3.77
Attention.K (6.82%)	Acc. nBERTs	87.7 ± 3.83 ∇ 80.01 ± 3.52 ∇	68.74 ± 2.29 ∇ 61.25 ± 2.07 ∇	65.48 ± 6.26 62.41 ± 6.5	87.8 ± 1.83 ∇ 79.55 ± 1.62 ∇	77.43 ± 3.55 70.8 ± 3.43
Attention.V (6.82%)	Acc. nBERTs	87.72 ± 3.16 ∇ 79.87 ± 2.92 ∇	67.22 ± 2.14 ∇ 60.12 ± 1.9 ∇	68.11 ± 5.19 ∇ 65.67 ± 5.09 ∇	88.17 ± 2.38 ∇ 79.63 ± 2.26 ∇	77.81 ± 3.22 71.32 ± 3.04
LayerNorm (0.02%)	Acc. nBERTs	21.37 ± 2.06 ∇ 0.0 ± 0.0 ∇	68.71 ± 1.89 ∇ 0.0 ± 0.0 ∇	0.29 ± 0.27 ∇ 0.24 ± 0.22 ∇	88.91 ± 1.74 ∇ 0.0 ± 0.0 ∇	44.82 ± 1.49 0.06 ± 0.06

Table 7: Summary of best performing SPARSEFIT configurations for T5-3B. We report the average and the standard deviation over the 60 few-shot train-validation splits for the **accuracy** metric and the normalized BERTScore (**nBERTs**). In brackets are the percentages of fine-tuned weights for each SPARSEFIT configuration. We show in **bold** the setting with the highest metric for each dataset. Significance testing was assessed via mean t-test in comparison with the baseline: ∇ represents a p-value lower than 10^{-2} .

that average performance, in terms of accuracy and NLE quality, do not increase when the rank is increased.

C.4 Explanations Shortcomings per Dataset

Given the diverse nature of the studied datasets, we perform an individual analysis for each dataset in order to find the particular deficiencies and traits of the explanations by dataset. Figure 22 shows a set of histograms with the assessment of the annotators on shortcomings for the e-SNLI dataset. It can be seen that the *Nonsensical* category is consistently the most common no matter what fine-tuning strategy was used. Below, the reader can find two examples of *Nonsensical* explanations generated by the *Baseline* and the *Decoder* strategy, respectively.

In addition to this, *Input Repetition* is the second most common shortcoming for e-SNLI. A regular pattern found in the generated explanations is the

repetition of a sub-string of the hypothesis as the predicted explanation, which happens for around 17% of the generated explanations. Below, the reader can see an example of input repetition found in the e-SNLI dataset.

We depict in Figure 25 a set of histograms with the number of times that a shortcoming category happens for different fine-tuning strategies for ECQA. Predominantly, *Incomplete Explanation* is the main weakness of generated NLEs. Notice that for this dataset, the answers are not generally shared by different samples (i.e., the possible labels for a sample are not always the same as in the other datasets). This causes the generated explanations to be vague and incomplete. Below, the reader can see 3 examples of *Incomplete Explanation* generated by the *Baseline*, *Decoder*, and *Encoder* fine-tuning strategy, respectively.

Figure 27 shows a set of histograms with the as-

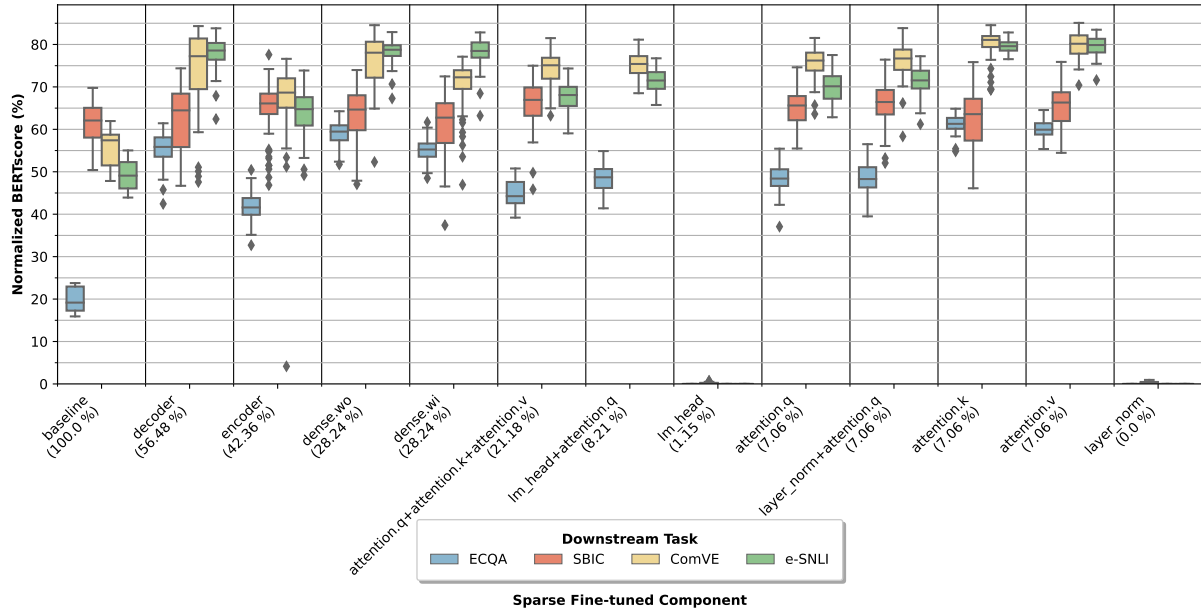


Figure 16: Distribution of the **normalized BERTScore** for different SPARSEFIT settings of sparse fine-tuning for T5-3b. The percentage of fine-tuned parameters is shown between brackets.

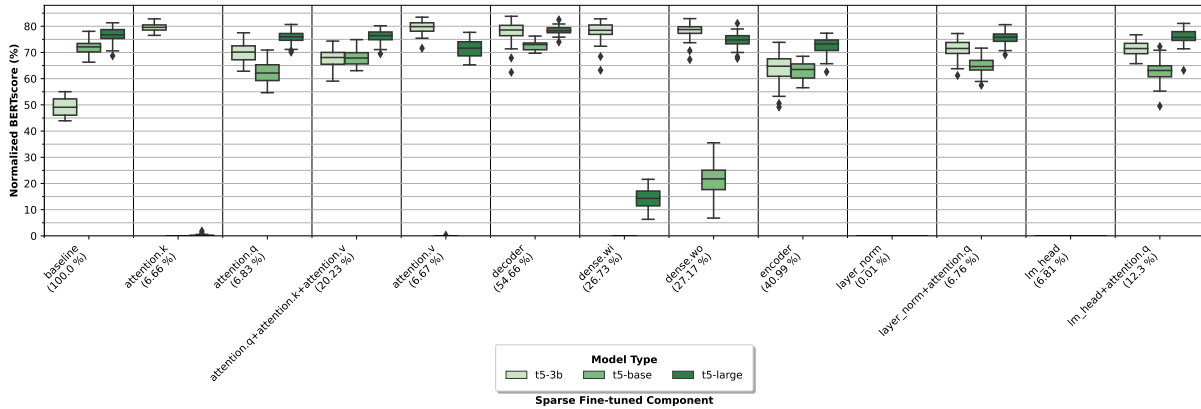


Figure 17: Distribution of the **normalized BERTScore** for different settings of sparse fine-tuning for the **e-SNLI** dataset. For each model, the box represents the overall performance over the 60 train-validation splits. The percentage of parameters fine-tuned for each setup is depicted in brackets below the name of each configuration.

assessment done by the annotators about the most common shortcomings. Different from other datasets, there is no singular shortcoming that dominates the results for all the fine-tuning setups. The most common mistakes among all the explanations in the dataset are: *Inaccurate*, *Nonsensical*, and *Incomplete Explanation*. Below, the reader can find an example for the *Incomplete Explanation* shortcoming for the *Decoder* fine-tuning.

We have depicted in Figure 28 a series of histograms with the frequency of possible shortcomings given by human annotators to the evaluated explanations. It can be seen that annotators consider that the *Lack of explanation*, *Nonsensical*, and *Incomplete Explanation* are the most relevant cate-

gories to explain the weaknesses of the generated explanations.

C.5 Inter-annotator Agreement

We show in Figure 30 an example of perceptual disagreement where the annotators assigned the same plausibility reason but a different score. Furthermore, Figure 31 shows an example of expectation disagreement where human evaluators assigned a opposite score for the given explanation.

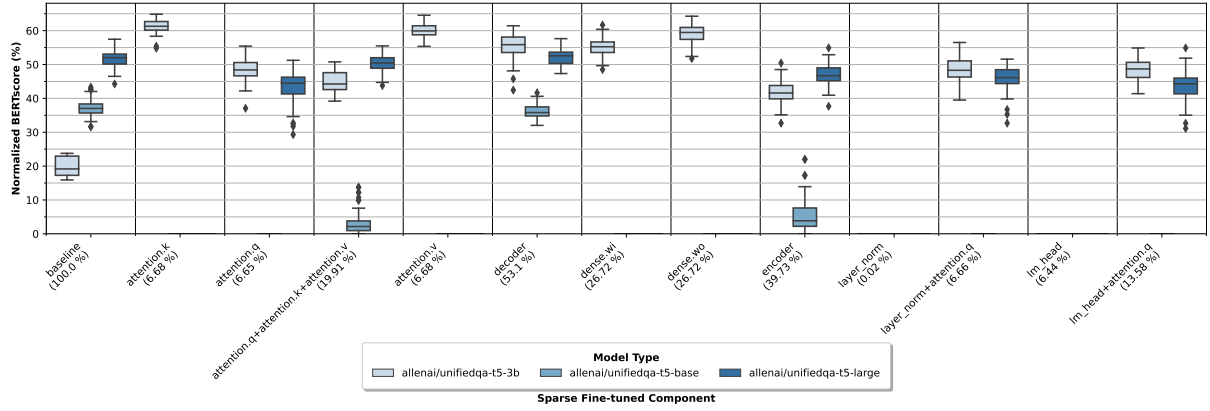


Figure 18: Distribution of the **normalized BERTscore** for different settings of sparse fine-tuning for the **ECQA** dataset. For each model, the box represents the overall performance over the 60 train-validation splits. The percentage of parameters fine-tuned for each setup is depicted in brackets below the name of each configuration.

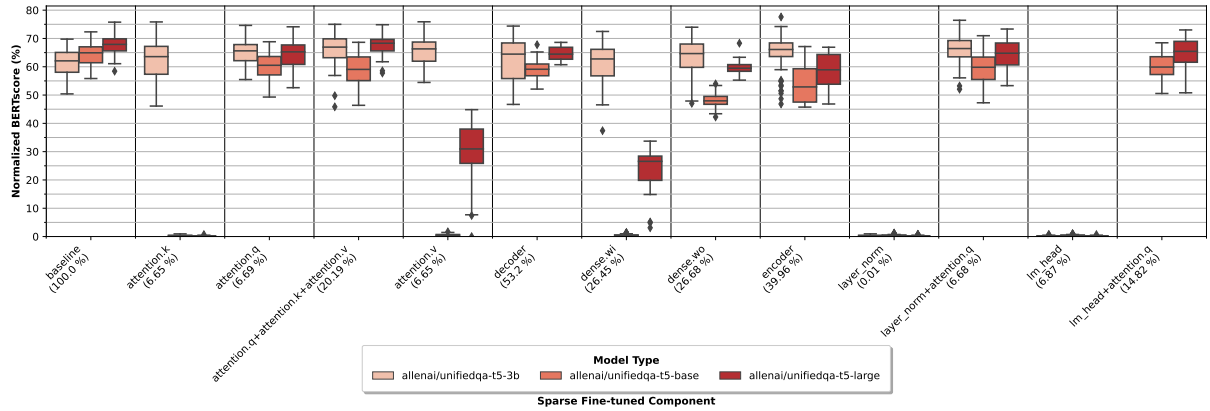


Figure 19: Distribution of the **normalized BERTscore** for different settings of sparse fine-tuning for the **SBIC** dataset. For each model, the box represents the overall performance over the 60 train-validation splits. The percentage of parameters fine-tuned for each setup is depicted in brackets below the name of each configuration.

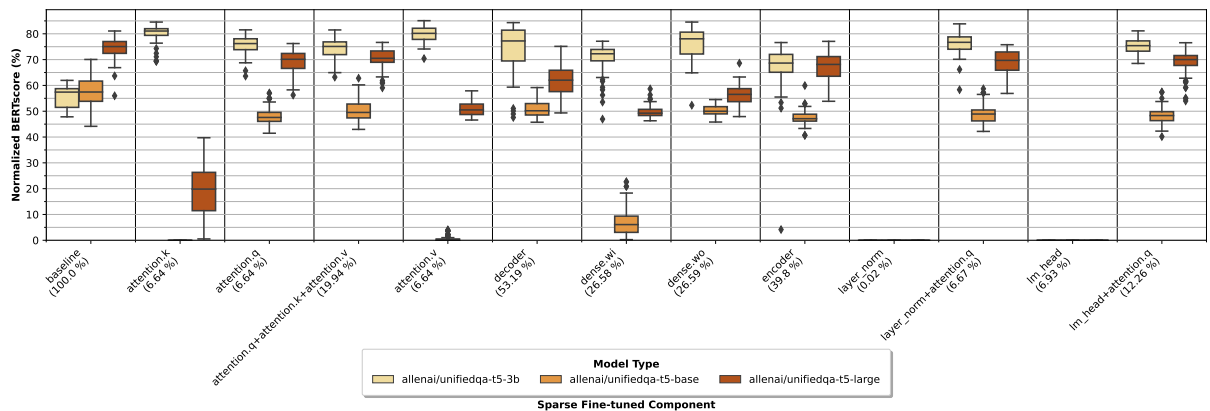


Figure 20: Distribution of the **normalized BERTscore** for different settings of sparse fine-tuning for the **ComVE** dataset. The baseline model represents the work done by (Marasovic et al., 2022), where all the parameters of the LM were fine-tuned. For each model, the box represents the overall performance over the 60 train-validation splits. The percentage of parameters fine-tuned for each setup is depicted in brackets below the name of each configuration.

PEFT Strategy	Rank Size	Percentage Parameters		ComVE	ECQA	SBIC	e-SNLI	Avg
LoRA	8	0.32%	Acc.	67.64 $\pm$ 3.37	39.59 $\pm$ 3.82	63.42 $\pm$ 3.46	84.15 $\pm$ 2.0	63.7 $\pm$ 3.16
			nBERTs	61.24 $\pm$ 3.09	1.55 $\pm$ 1.26	60.93 $\pm$ 3.45	76.41 $\pm$ 1.82	50.03 $\pm$ 2.4
	16	0.63%	Acc.	67.94 $\pm$ 3.4	39.41 $\pm$ 3.58	63.26 $\pm$ 3.36	84.26 $\pm$ 1.88	63.72 $\pm$ 3.06
			nBERTs	61.51 $\pm$ 3.11	1.44 $\pm$ 1.31	60.78 $\pm$ 3.49	76.5 $\pm$ 1.71	50.06 $\pm$ 2.41
	32	1.26%	Acc.	67.79 $\pm$ 3.75	39.74 $\pm$ 3.85	63.5 $\pm$ 3.28	84.27 $\pm$ 1.9	63.82 $\pm$ 3.2
			nBERTs	61.36 $\pm$ 3.43	1.36 $\pm$ 1.18	61.01 $\pm$ 3.36	76.51 $\pm$ 1.73	50.06 $\pm$ 2.43
	64	2.49%	Acc.	67.65 $\pm$ 3.77	43.44 $\pm$ 3.54	63.78 $\pm$ 3.15	84.25 $\pm$ 1.91	64.86 $\pm$ 3.11
			nBERTs	61.31 $\pm$ 3.42	0.32 $\pm$ 0.40	61.10 $\pm$ 3.31	76.54 $\pm$ 1.73	50.01 $\pm$ 2.10
	128	4.86%	Acc.	67.77 $\pm$ 3.73	43.51 $\pm$ 3.57	63.57 $\pm$ 3.16	84.26 $\pm$ 1.92	64.78 $\pm$ 3.1
			nBERTs	61.36 $\pm$ 3.41	0.33 $\pm$ 0.41	61.06 $\pm$ 3.29	76.49 $\pm$ 1.75	49.81 $\pm$ 2.22

Table 8: Accuracy and NLE quality metrics for different rank sizes in LoRA. We report the average and the standard deviation over the 60 few-shot train-validation splits for the **accuracy** metric and the normalized BERTScore (**nBERTs**).

Premise	A poor family is leaving their home with only a few belongings
Hypothesis	A man eats a chalupa
Label	Contradiction
Explanation	A family consists of two or more people, not just one man.
	Generated Explanation
	"a man who eats a chalupa also has to be poor"
Human Score	No
Reason	Nonsensical

Premise	A man in red pants skiing down a slope
Hypothesis	An Olympic skier skiing.
Label	Entailment
Explanation	WE have no idea if the man is an olympic skier or not.
	Generated Explanation
	"we don't know what he is doing"
Human Score	No
Reason	Nonsensical

Figure 21: Examples of **Non-sensical** NLEs generated for e-SNLI.

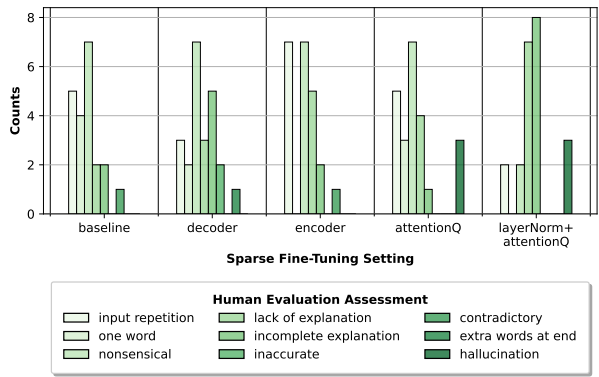


Figure 22: Histogram of the occurrences of the main shortcomings of the generated explanations for the baseline and the two best performing sparse fine-tuning setup for the **e-SNLI** dataset.

Premise	girl in uniform running through the water fountain gushing water from the tiles
Hypothesis	A girl is running through the water
Label	Entailment
Explanation	"If a water fountain is gushing water, that implies that there is water, and therefore the girl is running through water"
	Generated Explanation
	"a girl in uniform running through the water fountain."
Human Score	No
Reason	Input Repetition

Figure 23: Examples of **Input-Repetition** NLEs generated for e-SNLI.

Question	Where is the ticket booth often outside the building?
Choices	train station — stadium — auditorium — subway
Label	auditorium
Explanation	Ticket booth of an auditorium is often outside the building
	Generated Explanation
	"auditorium is a large venue with lots of people. the ticket booth is often outside the building"
Human Score	Weak Yes
Reason	Incomplete Explanation

Question	What happens to the person that is well known for frequent farting?
Choices	social ostracism — evil — bad smell — stinking
Label	social ostracism
Explanation	Ticket booth of an auditorium is often outside the building
	Generated Explanation
	"Social ostracism is act of ignoring and excluding individuals. Frequent farting can make you ignored by people around you as frequent farting makes them uncomfortable with you"
Human Score	Weak Yes
Reason	Incomplete Explanation

Figure 24: Examples of **Incomplete Explanation** NLEs generated for ECQA.

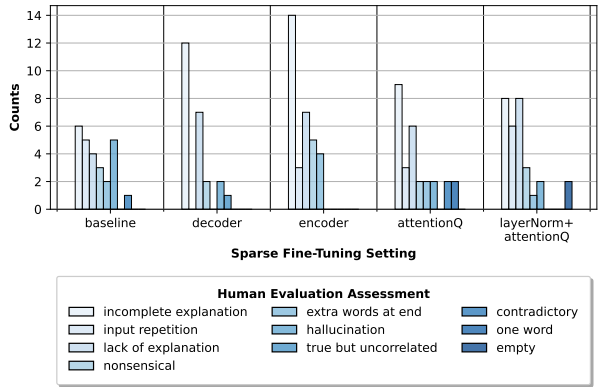


Figure 25: Histogram of the occurrences of the main shortcomings of the generated explanations for the baseline and the two best performing sparse fine-tuning setup for the **ECQA** dataset.

Post	The normie "vibe" argument is horseshit Anytime a normie says that you give off misogynistic vibes even if you don't outright say "I hate women" they are lying.
Label	Offensive
Explanation	this post is a personal attack
	Generated Explanation
	"this post implies that women are inferior"
Human Score	Weak No
Reason	Incomplete Explanation

Figure 26: Examples of **Incomplete Explanation** NLEs generated for SBIC.

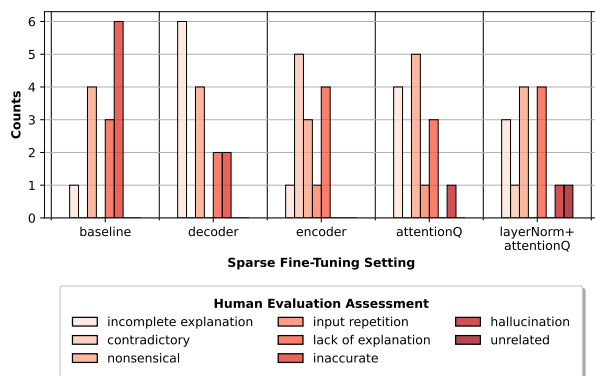


Figure 27: Histogram of the occurrences of the most common explanation shortcomings for the baseline and the two best performing sparse fine-tuning setup for the SBIC dataset.

Question	Teddy liked learning languages. He helped him with what?	
Choices	problems — frustration — confidence — better communication — sadness	
Label	better communication	
Explanation	Better communication is defined as verbal speech or other methods of relaying information that get a point across. He helped him with better communication.	
	Generated Explanation he helped him with better communication.	
	Annotator 1	Annotator 2
Human Score	Weak No	Weak Yes
Reason	Lack of Explanation	Lack of Explanation

Figure 30: Example of annotator perceptual disagreement in our study for the ECQA dataset.

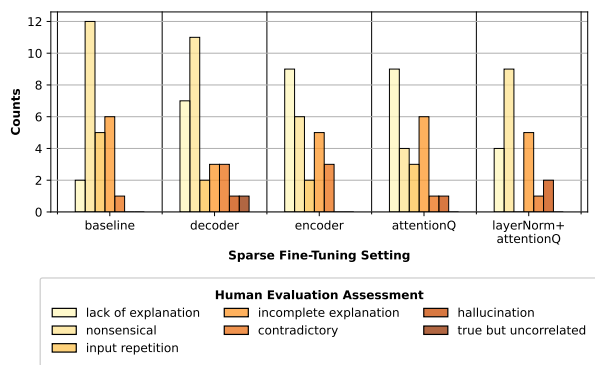


Figure 28: Histogram of the occurrences of the most common explanation shortcomings for the baseline and the two best performing sparse fine-tuning setup for the ComVE dataset.

Question	What is the best way to release energy?	
Choices	yell — think — exercise — rest — work off	
Label	exercise	
Explanation	By doing exercise, one can release energy. The best way of releasing energy is exercise.	
	Generated Explanation "exercise releases energy. to release energy, one must yell or yell loudly. to exercise, one must exercise vigorously. to work off, one must work off the energy."	
	Annotator 1	Annotator 2
Human Score	No	Weak Yes
Reason	Hallucination	Hallucination

Figure 29: Example of annotator expectation disagreement in our study for the ECQA dataset.

Question	What is the best way to release energy?	
Choices	yell — think — exercise — rest — work off	
Label	exercise	
Explanation	By doing exercise, one can release energy. The best way of releasing energy is exercise.	
	Generated Explanation "exercise releases energy. to release energy, one must yell or yell loudly. to exercise, one must exercise vigorously. to work off, one must work off the energy."	
	Annotator 1	Annotator 2
Human Score	No	Weak Yes
Reason	Hallucination	Hallucination

Figure 31: Example of annotator expectation disagreement in our study for the ECQA dataset.